WASP 2026 · ASTROCLIMB SHARED TASK · 4TH WORKSHOP ON AI FOR SCIENTIFIC PUBLICATIONS

# Modality-Conditioned Ensembles and Zero-Shot Cross-Encoder Reranking for Relation Classification Between Astronomy-Paper Objects

M. Tahsan Zahid · Most. Atkia Farzana · Tonmoy Sarker Sourav — Rajshahi University of Engineering and Technology (RUET), Bangladesh

**0.68759** best public LB pooled macro-F1

**0.2896** constraint-aware random floor

**17** leaderboard submissions

**0.908** best boundary AUC, text+text rel/unrel (w/ rerankers)

## 1 · THE TASK

Classify the relation between two objects (caption and/or figure) from astronomy papers into 4 classes: **same_figure, same_paper, related_papers, unrelated_papers.**

| SUBSET | TRAIN | TEST | CLASSES |
|---|---|---|---|
| image+text | 4,000 | 4,000 | all 4 |
| text+text | 3,000 | 3,000 | 3 |
| image+image | 3,000 | 3,000 | 3 |

**same_figure** is impossible outside image+text, so one unified classifier is structurally sub-optimal. Metric: **pooled macro-F1** over all 10,000 test rows — not the mean of the three subset scores.

## 2 · OUR APPROACH

**Encode** Qwen3-VL-Emb-2B

**Features** CSLS, hubness, TF-IDF

**Rerank** zero-shot, no FT

**Ensemble** GBM×2+LR, cascade

One independent ensemble per modality: two LightGBM models (scalar-feature + full-feature) plus logistic regression, blended, with a per-class threshold (coordinate ascent) and a binary related-vs-unrelated cascade. Shared **StratifiedKFold(5, shuffle=True, random_state=0)** OOF protocol keeps every stage blendable.

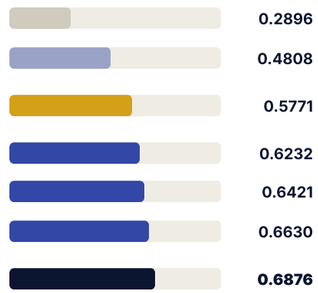
Per subset: **45 / 31 / 42** scalar pair features (img+txt / txt+txt / img+img) plus embedding-pair blocks (|e1−e2| & e1·e2 per space; 5,632/7,168/7,168 dims), for **5,677 / 7,199 / 7,210** inputs in the full block.

Scalar features: cosine & CSLS similarity, log-rank hubness, word/char n-gram (3–5) TF-IDF cosine, rare-token & entity overlap, 2-hop proxy similarity, text/OCR length, and reranker-derived features (raw score, percentile rank, deviation from mean, object degree).

**Rerankers:** Qwen3-VL-Reranker-2B (image+text, image+image) and Qwen3-Reranker-4B (text+text), each pair scored both orders and averaged under a task-instruction prompt.

*Standing rules: id never used (label-sorted CSVs leak it); solution.csv never used for training/tuning.*

## 3 · PUBLIC LEADERBOARD PROGRESSION




Per-subset macro-F1 of the final model (img+txt / txt+txt / img+img): **0.6407 / 0.7310 / 0.5847**. Six hand-engineered feature ideas each moved pooled macro-F1 by <0.01.

**Encoder first** One joint vision-language embedding model beat the combined effect of every feature idea tried (0.4808→0.6232).

**Rerank, zero-shot** Untrained cross-encoders added a second large gain (0.6421→0.6630), no task-specific training.

**Features last** Hand-built features plateaued; several measurably hurt. Representation was the stronger bottleneck.

## 6 · ERROR ANALYSIS & PER-CLASS RESULTS

| CLASS | P | R | F1 |
|---|---|---|---|
| same_figure | 0.894 | 0.808 | 0.849 |
| same_paper | 0.678 | 0.597 | 0.635 |
| related_papers | 0.501 | 0.485 | 0.493 |
| unrelated_papers | 0.642 | 0.760 | 0.696 |
| **pooled macro-F1** | **0.668** | | |

78% of OOF errors fall at exactly two class boundaries: **same_paper↔related_papers** (1,471 errors) and **related_papers↔unrelated_papers** (1,690 errors). same_figure is comparatively easy (F1 0.849) but not solved. An oracle fixing 60% of boundary errors projects pooled macro-F1 up to ≈0.71–0.74.

ORACLE: FIXING A FRACTION OF BOUNDARY ERRORS

| BOUNDARY | 20% | 40% | 60% |
|---|---|---|---|
| same_paper↔related | 0.654 | 0.679 | 0.704 |
| related↔unrelated | 0.657 | 0.684 | 0.710 |

## 4 · AN UNTRAINED RERANKER RIVALS THE PIPELINE




Boundary AUC (text+text, rel vs. unrel): 0.844→0.905 with reranker features. On image+image the reranker alone roughly ties the pipeline (0.509 vs 0.520); image+image stays weakest (AUC only 0.672/0.683) — a stronger image-only encoder looks like the next lever there.

| SUBSET/BOUNDARY | RERANK. | PIPELINE | BOTH |
|---|---|---|---|
| txt+txt sp/rel | 0.824 | 0.802 | 0.818 |
| txt+txt rel/unrel | 0.892 | 0.844 | 0.905 |
| img+img sp/rel | 0.710 | 0.672 | 0.708 |
| img+img rel/unrel | 0.653 | 0.685 | 0.693 |

At same_paper-vs-related, the reranker alone is already as good as the full ensemble (0.824 vs 0.818 txt+txt) — it largely **supplants, not complements**, the engineered similarity features at that boundary.

## 5 · DISTANT SUPERVISION & NEGATIVE RESULTS

**12× more data, lower score.** 120,000 pairs mined from the source corpus (94,059 figures, 9,998 papers, shared DOI/citation edges; 0.0% test-caption leakage, independently confirmed) reached 0.5771 vs 0.6232 for the encoder swap alone.

| HF TRANSFER METRIC | VALUE |
|---|---|
| Held-out macro-F1 | 0.5929 |
| Zero-shot→Kaggle text+text | 0.6242 |
| Zero-shot→Kaggle image+image | 0.4158 |

**Rejected ideas:** class-count matching (−0.0087); SigLIP added to text+text (−0.0109, 64-token truncation); MLP blend; 5-knob one-vs-rest (overfits vs 2-knob cascade); OCR as same_paper signal (chance AUC); label-transitivity propagation (anti-transitive in practice, $\chi^2$=59.5, p<1e-4).

| SIGLIP ABLATION (SUBSET) | ΔF1 |
|---|---|
| image+text (clip+sig) | +0.0387 |
| text+text (clip+sci+sig) | −0.0109 |

***Validation bug:*** *an unshuffled, label-sorted CSV broke our nested split (F1≈0.12 vs 0.60 on identical run). Fix: permute indices with a seeded RNG before splitting.*

**Structural wins:** per-class threshold via coordinate ascent (+0.011) and a 2-knob related-vs-unrelated cascade (+0.006 to +0.011) — the only model-side changes that reliably helped.

## CAVEATS

Blend weights, thresholds and cascade tuned on the same row-level OOF predictions (mildly optimistic); folds not paper-grouped; single seed (random_state=0); 17 leaderboard submissions, so ~0.01 differences are within noise; the 0.68759 configuration is not ablated. Two fine-tuning notebooks were not verified to completion and carry no claim.

## NEXT STEPS

- Task-specific cross-encoder fine-tuning
- Paper-grouped cross-validation
- A full reranker ablation (embedding-only, reranker-only, features-only factorial)
- Multi-seed confidence intervals
- Fixed-representation learning curve (10K/30K/60K/120K) for the distant-supervision branch
- Estimate distant-supervision label noise
- Trial transductive paper-clustering post-processing (7,922 test clusters found; shipped as unscored A/B variant, not in base score)

## ENGINEERING FINDINGS & BUILT ON

**GPU OOM fix:** calling a VL model's full generation head for hidden states materializes a 151,936-vocab logits tensor (212 MB per 731-token sequence) that's never used — call the inner transformer directly and read `last_hidden_state`.

**Caching:** a CPU stage decodes/resizes all images once to a shared 384px JPEG cache + parquet tables, persisted as a Kaggle Dataset; embeddings are deduplicated by content hash and L2-normalized on save.

**Reproducibility:** every result traces to a raw OOF array, notebook output, or the results log; identical StratifiedKFold(5) protocol across every notebook contributing an OOF feature. All models open-weight, run locally — no external LLM API.

**Built on:** CLIP, SigLIP, DINOv2, SciNCL encoders; LightGBM; Qwen3-VL-Embedding / Qwen3-VL-Reranker / Qwen3-Reranker (zero-shot); scikit-learn CV.

Code & notebooks in the public repository · WASP 2026 (AstroCLIMB), 4th Workshop on AI for Scientific Publications · source corpus: adsabs/AstroCLIMB (HuggingFace)